BadmintonAI - Product Requirements Document

1. Product Overview

1.1 Product Vision

BadmintonAI is an Android application that uses AI-powered pose estimation to analyze badminton players' techniques, provide real-time scoring, and deliver personalized coaching feedback to help players improve their skills.

1.2 Target Users

User Type	Description	Pain Points
Amateur Players	Recreational players wanting to improve	No access to professional coaching, can't see own form
Intermediate Players	Club-level players seeking technique refinement	Need objective feedback on specific strokes
Coaches	Badminton coaches training multiple students	Time-consuming to analyze each student's videos manually
Self-learners	Players learning from online tutorials	Can't verify if they're executing techniques correctly

1.3 Core Value Proposition

- **Accessible Coaching**: Professional-level technique analysis without hiring a coach
- **Objective Feedback**: AI-based scoring removes subjective bias
- **Instant Analysis**: Real-time or near-real-time feedback during practice
- **Progress Tracking**: Historical data to visualize improvement over time

2. Technical Feasibility Analysis

2.1 Pose Estimation Technology Options

Technology	Provider	Keypoints	Performance	Pros	Cons
MediaPipe Pose	Google	33 full-body	~30 FPS on mid-range	Cross-platform, 3D coords, open-source	Requires custom analysis layer
ML Kit Pose Detection	Google	33 landmarks	23-30 FPS	Easy Firebase integration, two accuracy modes	Limited customization
QuickPose SDK	QuickPose.ai	33 points	Optimized for NPU	Pre-built sports features, rep counting	Commercial licensing
TensorFlow Lite (MoveNet)	Google	17 keypoints	~30 FPS	Lightweight, proven accuracy	Fewer keypoints than alternatives

Recommendation: MediaPipe Pose as primary solution

- Full 33-point skeleton including hands (important for grip analysis)
- 3D world coordinates enable depth-aware analysis
- Apache 2.0 license, no cloud dependency
- Strong community support and documentation

2.2 Academic Research Insights

Recent research (2024-2025) validates the feasibility:

- **Nature Scientific Reports (2025)**: Machine learning models can classify badminton stroke types with high accuracy using pose keypoints

- **BST Transformer**: State-of-the-art stroke classification using skeleton data + shuttlecock trajectory
- **MultiSenseBadminton Dataset**: Comprehensive dataset with 7,763 swings across skill levels proves biomechanical analysis is viable
- **AI-Powered Training Systems**: Frame-by-frame correction systems using MediaPipe are already demonstrated

3. Functional Requirements

3.1 Core Features (MVP - Phase 1)

F1: Video Recording & Import

ID	Requirement	Priority
F1.1	Record video using device camera (rear/front)	P0
F1.2	Import existing videos from gallery	P0
F1.3	Support video formats: MP4, MOV, AVI	P0
F1.4	Recommended recording guidance (distance, angle, lighting)	P1

F2: Pose Detection & Tracking

ID	Requirement	Priority
F2.1	Detect 33 body landmarks using MediaPipe Pose	P0
F2.2	Track pose across video frames	P0
F2.3	Handle single-player detection	P0
F2.4	Output 3D world coordinates for depth analysis	P1
F2.5	Skeleton overlay visualization on video	P0

F3: Stroke Recognition

ID	Requirement	Priority
F3.1	Identify stroke type from motion sequence	P0
F3.2	Supported strokes (MVP): Forehand Clear, Backhand Clear, Smash, Drop Shot, Serve	P0
F3.3	Auto-detect stroke start/end frames	P1
F3.4	Manual stroke marking as fallback	P1

Supported Stroke Types:

```
Phase 1 (MVP):
├── Overhead Strokes
│   ├── Forehand Clear (高远球)
│   ├── Smash (杀球)
│   └── Drop Shot (吊球)
├── Underhand Strokes
│   └── Serve (发球)
└── Net Play
    └── Net Shot (网前球)

Phase 2 (Future):
├── Backhand Clear (反手高远球)
├── Drive (平抽球)
├── Lift (挑球)
└── Footwork Analysis (步法分析)
```

F4: Technique Scoring System

ID	Requirement	Priority
F4.1	Generate overall score (0-100) for each stroke	P0
F4.2	Provide sub-scores for key technique dimensions	P0
F4.3	Compare against reference "ideal form" models	P0
F4.4	Display score breakdown with visual indicators	P0

Scoring Dimensions:

Dimension	Weight	Metrics Analyzed
Preparation	20%	Ready position, racket preparation angle, split step
Backswing	15%	Elbow height, shoulder rotation, weight transfer
Contact Point	25%	Arm extension, contact height, body alignment
Follow-through	15%	Swing completion, balance maintenance
Timing & Rhythm	15%	Movement fluidity, acceleration curve
Footwork	10%	Base position, recovery step

F5: Coaching Feedback

ID	Requirement	Priority
F5.1	Generate text-based feedback for each scoring dimension	P0
F5.2	Highlight specific improvement areas	P0
F5.3	Provide actionable correction suggestions	P0
F5.4	Show visual comparison: user vs ideal form	P1
F5.5	Support feedback in Chinese (Simplified)	P0

Feedback Examples:

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Score: 72/100 - Forehand Clear

✓ Good: Preparation stance is solid, weight on back foot
⚠ Improve: Contact point is too low (shoulder height detected, should be above head)
⚠ Improve: Elbow drops before contact - maintain high elbow position
✗ Issue: Follow-through incomplete - extend arm fully toward target

Drill Suggestion: Practice shadow swings focusing on high contact point
```

3.2 Extended Features (Phase 2)

F6: Progress Tracking

ID	Requirement	Priority
F6.1	Store historical analysis results	P1
F6.2	Display score trends over time (charts)	P1
F6.3	Track improvement per stroke type	P1
F6.4	Set personal goals and track progress	P2

F7: Real-time Analysis Mode

ID	Requirement	Priority
F7.1	Live camera preview with pose overlay	P1
F7.2	Real-time audio/visual feedback during practice	P2
F7.3	Session recording with analysis	P2

F8: Social & Sharing

ID	Requirement	Priority
F8.1	Export analysis results as video/image	P2
F8.2	Share to social platforms	P2
F8.3	Compare with friends	P3

4. Non-Functional Requirements

4.1 Performance

Metric	Target	Notes
Video analysis speed	≤2x real-time	10s video analyzed in ≤20s

Metric	Target	Notes
Pose detection FPS	≥20 FPS	On mid-range devices (Snapdragon 6-series)
App startup time	<3s	Cold start to main screen
Memory usage	<500MB	During video analysis

4.2 Device Compatibility

Requirement	Specification
Minimum Android version	Android 8.0 (API 26)
Target Android version	Android 14 (API 34)
Minimum RAM	4GB
Camera	Required (for recording feature)
GPU	Recommended for real-time analysis

4.3 Privacy & Data

Requirement	Description
On-device processing	All pose analysis runs locally, no cloud upload
Video storage	User controls video retention
No account required	Core features work without login
GDPR/Privacy compliance	Clear data handling disclosure

4.4 Localization

Language	Priority
Simplified Chinese	P0
English	P1

5. Technical Architecture

5.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────┐
│                      Presentation Layer                  │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐  │
│  │  Camera  │  │  Video   │  │ Analysis │  │ Progress │  │
│  │  Screen  │  │  Player  │  │  Results │  │  Charts  │  │
│  └──────────┘  └──────────┘  └──────────┘  └──────────┘  │
└──────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────┐
│                      Domain Layer                        │
│  ┌──────────────────┐  ┌──────────────────┐              │
│  │  Stroke Analysis │  │  Scoring Engine  │              │
│  │     Service      │  │                  │              │
│  └──────────────────┘  └──────────────────┘              │
│  ┌──────────────────┐  ┌──────────────────┐              │
│  │ Feedback Generator│  │ Progress Tracker │              │
│  └──────────────────┘  └──────────────────┘              │
└──────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────┐
│                        ML Layer                          │
│  ┌──────────────────┐  ┌──────────────────┐              │
│  │   MediaPipe      │  │  Stroke Type     │              │
│  │   Pose Detector  │  │  Classifier (ML) │              │
│  └──────────────────┘  └──────────────────┘              │
│  ┌──────────────────┐  ┌──────────────────┐              │
│  │  Reference Pose  │  │  Angle/Distance  │              │
│  │     Models       │  │   Calculator     │              │
│  └──────────────────┘  └──────────────────┘              │
└──────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────┐
│                       Data Layer                         │
│  ┌──────────────────┐  ┌──────────────────┐              │
│  │   Room Database  │  │  Video File      │              │
│  │  (Analysis Logs) │  │   Storage        │              │
```

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│   └──────────────────┘   └──────────────────┘                    │
└──────────────────────────────────────────────────────────────────┘
```

5.2 Technology Stack

Layer	Technology	Rationale
UI Framework	Jetpack Compose	Modern declarative UI, better performance
Architecture	MVVM + Clean Architecture	Separation of concerns, testability
DI	Hilt	Official Android DI, integrates with Jetpack
Pose Detection	MediaPipe Pose (Android SDK)	Best balance of accuracy/performance
Video Processing	CameraX + MediaCodec	Modern camera API, hardware decoding
Local Storage	Room + DataStore	Structured data + preferences
Async	Kotlin Coroutines + Flow	Native Kotlin async, lifecycle-aware
ML Runtime	TensorFlow Lite	Custom stroke classifier model
Charts	MPAndroidChart or Vico	Progress visualization

5.3 Pose Analysis Pipeline

```
Video Frame
     │
     ▼
┌─────────────────┐
│  MediaPipe Pose │ ──► 33 Landmarks (x, y, z, visibility)
│    Detector     │
└─────────────────┘
     │
     ▼
┌─────────────────┐
│ Landmark Filter │ ──► Smooth landmarks, remove noise
│  & Smoother     │
└─────────────────┘
     │
     ▼
┌─────────────────┐
│  Angle/Vector   │ ──► Joint angles, body segment vectors
│   Calculator    │
└─────────────────┘
     │
     ▼
┌─────────────────┐
│  Stroke Phase   │ ──► Preparation → Backswing → Contact → Follow-through
│    Detector     │
└─────────────────┘
     │
     ▼
┌─────────────────┐
│ Reference Model │ ──► Compare against ideal form
│   Comparator    │
└─────────────────┘
     │
     ▼
┌─────────────────┐
│  Score & Report │ ──► Numerical scores + text feedback
│    Generator    │
└─────────────────┘
```

5.4 Key Biomechanical Metrics

Metric	Landmarks Used	Ideal Range (Forehand Clear)
Elbow Angle	Shoulder-Elbow-Wrist	90-120° at backswing, 150-180° at contact
Shoulder Rotation	Hip-Shoulder vector angle	45-90° rotation from ready position
Contact Height	Wrist Y relative to head	Above head level (ratio > 1.1)
Trunk Rotation	Hip-Shoulder alignment	30-60° rotation at contact
Weight Transfer	Hip center trajectory	Forward shift during stroke
Arm Extension	Shoulder-Wrist distance	Near full extension at contact

6. UI/UX Design Guidelines

6.1 Key Screens

Screen	Purpose	Key Components
Home	Entry point, quick actions	Record button, Import button, Recent analyses
Recording	Capture video	Camera preview, recording guides, timer
Analysis Loading	Processing feedback	Progress indicator, current stage display
Results	Score & feedback display	Score card, video playback with overlay, detailed breakdown
History	Past analyses	List view, filters by stroke type, date range
Progress	Trend visualization	Charts, milestone badges, improvement highlights

6.2 UI Prototypes

> Interactive HTML prototypes: `docs/ui-prototypes/*.html`

Home Screen (首页)

- Main entry with recording/import actions
- Recent analyses with score badges
- Bottom navigation bar

Recording Screen (录制)

- Camera preview with pose guide silhouette
- Stroke type selector
- Recording tips and positioning guidance

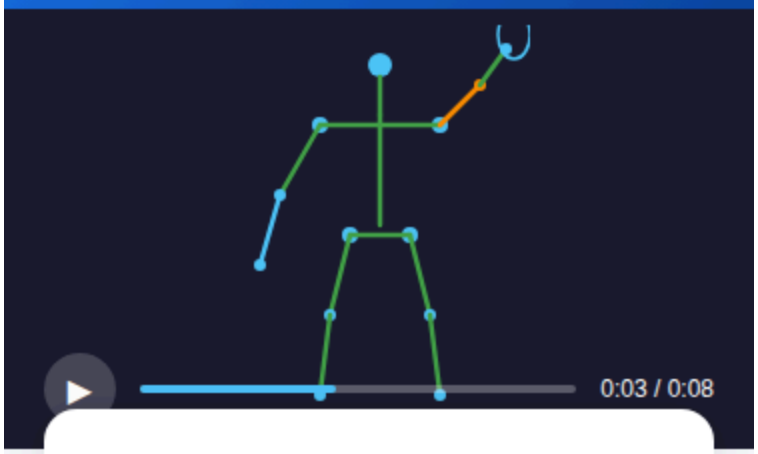
Analyzing Screen (分析中)

- Skeleton animation visualization
- Progress steps indicator
- Processing status with ETA

Results Screen (分析结果)

分析结果

正手高远球 · 2024年3月9日



78
综合评分

良好水平

✓ 超过 65% 的用户

88 准备动作	75 引拍	62 击球点
80 随挥	82 节奏	78 步法

详细反馈

✓ **准备姿势标准**
重心稳定，双脚站位合理，随时准备移动

! **击球点偏低**
击球时手臂未完全伸展，击球点在肩部高度。建议在头顶上方击球，可获得更大力量

! **引拍时肘部下沉**
引拍阶段肘部角度约85°，理想角度应为100-120°。保持高肘位可增加挥拍行程

✓ **随挥完整**
击球后手臂顺势下摆，动作流畅自然

推荐练习

高点击球专项练习
对墙挥拍练习，专注于在最高点触球

- Overall score with grade
- 6-dimension sub-scores
- Skeleton overlay with problem areas highlighted (orange)
- Detailed text feedback with ✓/⚠ indicators
- Recommended drill suggestions

History Screen (历史记录)

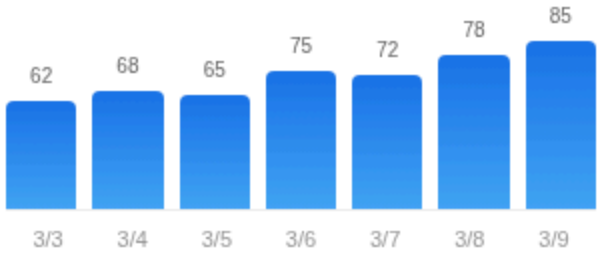
训练记录

追踪你的进步轨迹

27	76	+12
本月练习	平均得分	月进步

得分趋势

正手高远球



↑ 较上周提升 15%，击球点维度进步最大

全部 | 正手高远球 | 杀球 | 吊球 | 发球

历史记录

今天

正手高远球 +7

14:32 · 时长 8秒

85

首页 | 进步 | 历史 | 设置

昨天

正手高远球

16:45 · 时长 7秒

78

吊球 -3

- Stats summary (monthly practice, average score, improvement)
- Score trend chart with stroke filter
- Grouped history list with score badges and trends

6.3 Design Principles

1. **Clarity First**: Technical metrics presented in understandable terms
2. **Visual Feedback**: Use skeleton overlay, color-coded scoring
3. **Actionable**: Every critique paired with improvement suggestion
4. **Encouraging**: Celebrate progress, avoid demotivating language

6.4 Color Coding

Score Range	Color	Meaning
90-100	Green	Excellent technique
70-89	Blue	Good, minor improvements needed
50-69	Yellow	Fair, specific areas to work on
0-49	Red	Needs significant improvement

7. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
Pose detection inaccuracy in fast movements	High	Medium	Use temporal smoothing, require proper video setup
Varied camera angles affect analysis	High	High	Provide strict recording guidelines, detect bad angles
Different body types affect ideal metrics	Medium	Medium	Parameterize reference models, allow calibration
User frustration with low scores	Medium	Medium	Focus on improvement suggestions, show progress
Device performance variance	Medium	High	Provide quality/speed toggle, test on low-end devices

8. Success Metrics

Metric	Target	Measurement
DAU/MAU	>30%	Analytics
Analysis completion rate	>80%	Users who complete full analysis flow
Repeat usage	>3 analyses per user per week	Analytics
User-reported improvement	>60% feel improvement after 4 weeks	In-app survey
App Store rating	>4.2 stars	Store metrics

9. Project Phases & Timeline

Phase 1: MVP (8-10 weeks)

- Core pose detection integration
- 5 stroke type analysis
- Basic scoring system
- Text feedback generation
- Video recording & import

Phase 2: Enhancement (6-8 weeks)

- Progress tracking & history
- Visual comparison features
- Real-time preview mode
- Performance optimization

Phase 3: Growth (4-6 weeks)

- Social sharing features
- Additional stroke types
- Coach/student features
- Localization expansion

10. Open Questions

Question	Owner	Due Date
Which stroke types to prioritize for MVP?	Product	Week 1
Minimum acceptable pose detection accuracy?	Engineering	Week 2
Reference models: use existing dataset or create custom?	ML Team	Week 2
Monetization strategy (free/freemium/paid)?	Business	Week 3
Need professional coach consultation for scoring rubric?	Product	Week 2

Appendix A: Competitive Analysis

App	Strengths	Weaknesses
HomeCourt (Basketball)	Real-time tracking, polished UX	Basketball only, subscription model
SwingVision (Tennis)	Court detection, shot tracking	Tennis focused, requires Apple Watch
OnForm	Multi-sport, coach collaboration	Manual annotation required
Badminton Apps (current)	Score tracking, drill videos	No AI analysis, passive content only

Opportunity: No dedicated AI-powered badminton technique analysis app exists in the market.

Appendix B: Reference Pose Data Sources

1. **MultiSenseBadminton Dataset** - 7,763 swings with skill level annotations
2. **ShuttleSet** - Broadcast match data with stroke annotations
3. **Professional Coach Videos** - To be collected for reference models

Document Version: 1.0
Created: March 2026
Last Updated: March 9, 2026